Now hold Ae fixed and let B and C meet A; the angle cAg between curve and tangent vanishes, exactly as in Lemma VII, so the curvilinear areas Abd and Ace coincide with the rectilinear triangles Afd and Age. But these two triangles share the same vertex angle at A along the tangent Ag, so by Lemma V (similar rectilinear figures) their areas are as the squares of the homologous sides, that is $Afd : Age = Ad^2 : Ae^2$. Since $Ad : Ae = AD : AE$ and the original triangles $\triangle ABD$, $\triangle ACE$ stay always proportional to these blown-up areas, the same duplicate ratio descends to them. Therefore the areas $\triangle ABD$ and $\triangle ACE$ are ultimately one to the other as $AD^2 : AE^2$. *Q.E.D.*